Time Series Analysis

Part 1: Lecture 3

ARIMA Models

Outline

1 Autoregressive (AR) & Moving Average (MA) models

2 AR(I)MA models

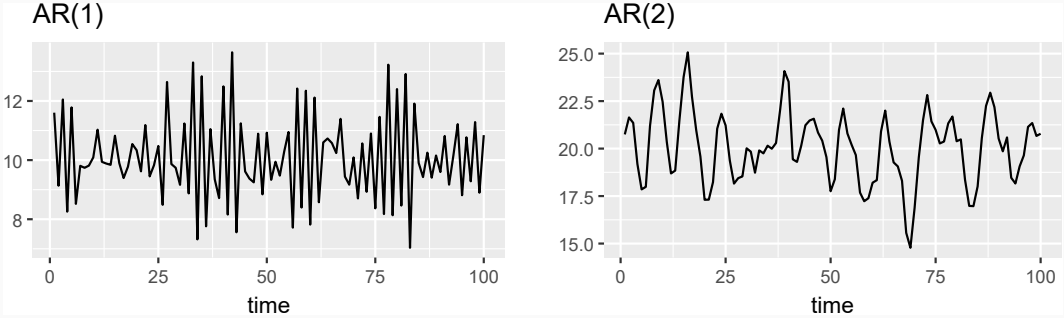
3 Seasonal ARIMA models

Autoregressive (AR) models

Autoregressive (AR) models:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t,$$

where ε_t is white noise. A multiple regression with **lagged values** of y_t as predictors.



- Cyclic behaviour is possible when $p \geq 2$.

Autoregressive (AR) models

Stationarity conditions

We normally restrict autoregressive models to stationary data, and then some constraints on the values of the parameters are required.

General condition for stationarity

Complex roots of $1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p$ lie outside the unit circle on the complex plane.

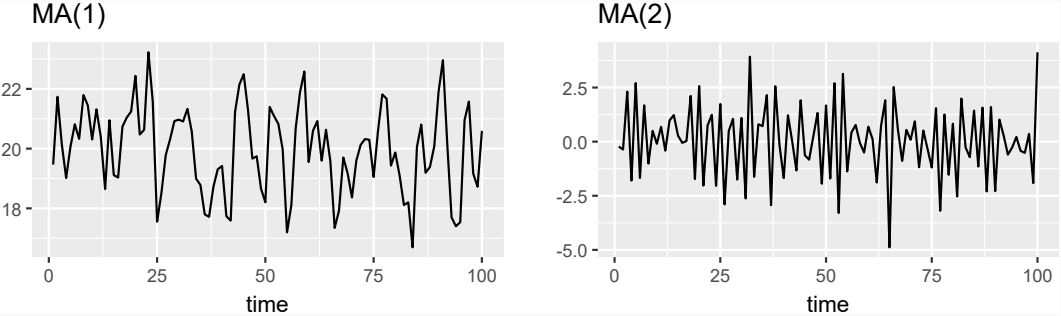
- For $p = 1$: $-1 < \phi_1 < 1$.
- For $p = 2$:
  $-1 < \phi_2 < 1 \qquad \phi_2 + \phi_1 < 1 \qquad \phi_2 - \phi_1 < 1.$
- More complicated conditions hold for $p \geq 3$.
- Estimation software takes care of this.

Moving Average (MA) models

Moving Average (MA) models:

$$y_t = c + \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \cdots + \theta_q\varepsilon_{t-q},$$

where ε_t is white noise. A multiple regression with **lagged *errors*** as predictors.
Don't confuse with moving average smoothing!



Moving Average (MA) models

Invertibility

- Any MA(q) process can be written as an AR(∞) process if we impose some constraints on the MA parameters.
- Then the MA model is called “invertible”.
- Invertible models have some mathematical properties that make them easier to use in practice.

Invertibility

General condition for invertibility

Complex roots of $1 + \theta_1 z + \theta_2 z^2 + \cdots + \theta_q z^q$ lie outside the unit circle on the complex plane.

- For $q = 1$: $-1 < \theta_1 < 1$.
- For $q = 2$: $-1 < \theta_2 < 1 \qquad \theta_2 + \theta_1 > -1 \qquad \theta_1 - \theta_2 < 1$.
- More complicated conditions hold for $q \geq 3$.
- Estimation software takes care of this.

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1 Autoregressive (AR) & Moving Average (MA) models

2 AR(I)MA models

3 Seasonal ARIMA models

ARIMA models

AR: autoregressive (lagged observations as inputs)
I: integrated (differencing to make series stationary)
MA: moving average (lagged errors as inputs)

An ARIMA model is rarely interpretable in terms of visible data structures like trend and seasonality. But it can capture a huge range of time series patterns.

Autoregressive Moving Average models:

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t.$$

- Predictors include both **lagged values of y_t and lagged errors.**

Autoregressive Integrated Moving Average models * Combine ARMA model with **differencing**. * d-differenced series follows an ARMA model. * Need to choose p, d, q and whether or not to include c.

ARIMA models

ARIMA(p, d, q) model

AR: p = order of the autoregressive part
I: d = degree of first differencing involved
MA: q = order of the moving average part.

- White noise model: ARIMA(0,0,0)
- Random walk: ARIMA(0,1,0) with no constant
- Random walk with drift: ARIMA(0,1,0) with const.
- AR(p): ARIMA(p,0,0)
- MA(q): ARIMA(0,0,q)

Backshift notation for ARIMA

- ARMA model:

$$y_t = c + \phi_1 B y_t + \cdots + \phi_p B^p y_t + \varepsilon_t + \theta_1 B \varepsilon_t + \cdots + \theta_q B^q \varepsilon_t$$

$$\text{or} \quad (1 - \phi_1 B - \cdots - \phi_p B^p) y_t = c + (1 + \theta_1 B + \cdots + \theta_q B^q) \varepsilon_t$$

- ARIMA(1,1,1) model:

$$(1 - \phi_1 B) \quad (1 - B) y_t \quad = \quad c + (1 + \theta_1 B) \varepsilon_t$$

$(1 - \phi_1 B)$: AR(1)

$(1 - B) y_t$: First difference

$(1 + \theta_1 B) \varepsilon_t$: MA(1)

Expand: $y_t = c + y_{t-1} + \phi_1 y_{t-1} - \phi_1 y_{t-2} + \theta_1 \varepsilon_{t-1} + \varepsilon_t$

Understanding ARIMA models

- If $c = 0$ and $d = 0$, the long-term forecasts will go to zero.
- If $c = 0$ and $d = 1$, the long-term forecasts will go to a non-zero constant.
- If $c = 0$ and $d = 2$, the long-term forecasts will follow a straight line.
- If $c \neq 0$ and $d = 0$, the long-term forecasts will go to the mean of the data.
- If $c \neq 0$ and $d = 1$, the long-term forecasts will follow a straight line.
- If $c \neq 0$ and $d = 2$, the long-term forecasts will follow a quadratic trend.

Forecast variance and *d* * The higher the value of *d*, the more rapidly the prediction intervals increase in size. * For *d* = 0, the long-term forecast standard deviation will go to the standard deviation of the historical data.

Partial autocorrelations

Partial autocorrelations measure relationship between y_t and y_{t-k}, when the effects of other time lags — $1, 2, 3, \ldots, k-1$ — are removed.

α_k = kth partial autocorrelation coefficient

= equal to the estimate of ϕ_k in regression:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_k y_{t-k} + \varepsilon_t.$$

- Varying number of terms on RHS gives α_k for different values of k.
- $\alpha_1 = \rho_1$
- same critical values of $\pm 1.96/\sqrt{T}$ as for ACF.
- Last significant α_k indicates the order of an AR model.

Using ACF & PACF for model ID

Model	ACF	PACF
AR(p)	Tails off slowly	Cuts off after lag p
MA(q)	Cuts off after lag q	Tails off slowly
ARMA(p,q)	Tails off slowly	Tails off slowly

Maximum likelihood estimation

Having identified the model order, we need to estimate the parameters c, $\phi_1, \ldots, \phi_p$, $\theta_1, \ldots, \theta_q$.

- MLE is very similar to least squares estimation obtained by minimizing

$$\sum_{t-1}^{T} e_t^2$$

- The `ARIMA()` function allows CLS or MLE estimation.
- Non-linear optimization must be used in either case.
- Different software will give different estimates.

Information criteria

Akaike's Information Criterion (AIC):

$$\text{AIC} = -2\log(L) + 2(p+q+k+1),$$

where L is the likelihood of the data, $k = 1$ if $c \neq 0$ and $k = 0$ if $c = 0$.

$$\text{AICc} = -2\log(L) + 2(p+q+k+1)\left[1 + \frac{(p+q+k+2)}{T-p-q-k-2}\right]$$

where L is the maximised likelihood fitted to the *differenced* data, $k = 1$ if $c \neq 0$ and $k = 0$ otherwise.

Bayesian Information Criterion:

$$\text{BIC} = \text{AIC} + [\log(T) - 2](p+q+k+1).$$

- Good models are obtained by minimizing either the AIC, AICc or BIC.
- preference is to use the AICc. **Note**: Cannot compare AICc for different values of d.

Traditional modelling procedure for ARIMA models

1. Plot the data. Identify any unusual observations.
2. If necessary, transform the data (using a Box-Cox transformation) to stabilize the variance.
3. If the data are non-stationary: take first differences of the data until the data are stationary.
4. Examine the ACF/PACF: Is an AR(p) or MA(q) model appropriate?
5. Try your chosen model(s), and use the AICc to search for a better model.
6. Check the residuals from your chosen model by plotting the ACF of the residuals, and doing a portmanteau test of the residuals. If they do not look like white noise, try a modified model.
7. Once the residuals look like white noise, calculate forecasts.

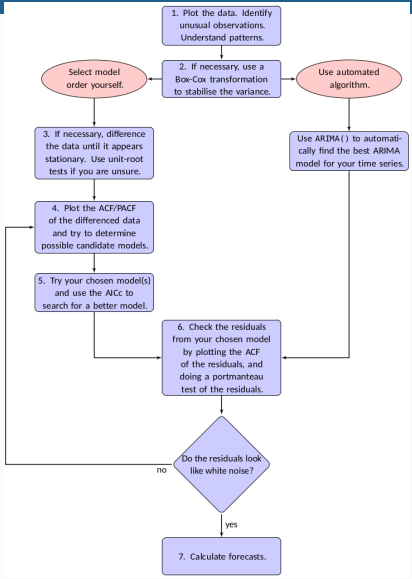
Automatic modelling procedure with `ARIMA()`

1. Plot the data. Identify any unusual observations.
2. If necessary, transform the data (using a Box-Cox transformation) to stabilize the variance.

3. Use `ARIMA` to automatically select a model.

6. Check the residuals from your chosen model by plotting the ACF of the residuals, and doing a portmanteau test of the residuals. If they do not look like white noise, try a modified model.
7. Once the residuals look like white noise, calculate forecasts.

Modelling procedure



How does ARIMA() work?

A non-seasonal ARIMA process

$$\phi(B)(1-B)^d y_t = c + \theta(B)\varepsilon_t$$

Need to select appropriate orders d, p, q, and whether to include the intercept c.

Hyndman and Khandakar (JSS, 2008) algorithm:

- Select no. differences d via KPSS test.
- Select p, q and inclusion of c by minimising AICc.
- Use stepwise search to traverse model space.

How does ARIMA() work?

Step1: Select current model (with smallest AICc) from:
ARIMA$(2, d, 2)$
ARIMA$(0, d, 0)$
ARIMA$(1, d, 0)$
ARIMA$(0, d, 1)$

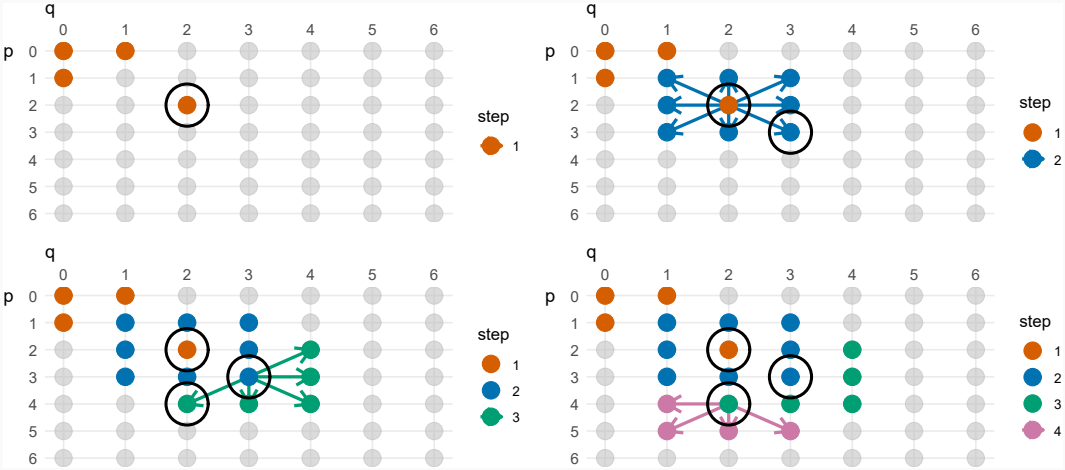
Step 2: Consider variations of current model:

- vary one of $p, q,$ from current model by ± 1;
- p, q both vary from current model by ± 1;
- Include/exclude c from current model.

Model with lowest AICc becomes current model.

Repeat Step 2 until no lower AICc can be found.

How does ARIMA() work?



Point forecasts

1. Rearrange ARIMA equation so y_t is on LHS.
2. Rewrite equation by replacing t by $T + h$.
3. On RHS, replace future observations by their forecasts, future errors by zero, and past errors by corresponding residuals.

Start with $h = 1$. Repeat for $h = 2, 3, \ldots$.

ARIMA(3,1,1) forecasts: Step 1

$$(1-\hat{\phi}_1B-\hat{\phi}_2B^2-\hat{\phi}_3B^3)(1-B)y_t = (1+\hat{\theta}_1B)\varepsilon_t,$$

$$\left[1-\hat{\phi}_1B-\hat{\phi}_2B^2-\hat{\phi}_3B^3-B+\hat{\phi}_1B^2+\hat{\phi}_2B^3+\hat{\phi}_3B^4\right] y_t = (1+\hat{\theta}_1B)\varepsilon_t$$

$$\left[1-(1+\hat{\phi}_1)B+(\hat{\phi}_1-\hat{\phi}_2)B^2+(\hat{\phi}_2-\hat{\phi}_3)B^3+\hat{\phi}_3B^4\right] y_t = (1+\hat{\theta}_1B)\varepsilon_t$$

$$y_t-(1+\hat{\phi}_1)y_{t-1}+(\hat{\phi}_1-\hat{\phi}_2)y_{t-2}+(\hat{\phi}_2-\hat{\phi}_3)y_{t-3}+\hat{\phi}_3y_{t-4} = \varepsilon_t+\hat{\theta}_1\varepsilon_{t-1}$$

$$y_t = (1+\hat{\phi}_1)y_{t-1}-(\hat{\phi}_1-\hat{\phi}_2)y_{t-2}-(\hat{\phi}_2-\hat{\phi}_3)y_{t-3}-\hat{\phi}_3y_{t-4}+\varepsilon_t+\hat{\theta}_1\varepsilon_{t-1}$$

Point forecasts (h=1)

$$y_t = (1 + \hat{\phi}_1)y_{t-1} - (\hat{\phi}_1 - \hat{\phi}_2)y_{t-2} - (\hat{\phi}_2 - \hat{\phi}_3)y_{t-3} - \hat{\phi}_3 y_{t-4} + \varepsilon_t + \hat{\theta}_1\varepsilon_{t-1}$$

ARIMA(3,1,1) forecasts: Step 2

$$y_{T+1} = (1 + \hat{\phi}_1)y_T - (\hat{\phi}_1 - \hat{\phi}_2)y_{T-1} - (\hat{\phi}_2 - \hat{\phi}_3)y_{T-2} - \hat{\phi}_3 y_{T-3} + \varepsilon_{T+1} + \hat{\theta}_1\varepsilon_T$$

ARIMA(3,1,1) forecasts: Step 3

$$\hat{y}_{T+1|T} = (1 + \hat{\phi}_1)y_T - (\hat{\phi}_1 - \hat{\phi}_2)y_{T-1} - (\hat{\phi}_2 - \hat{\phi}_3)y_{T-2} - \hat{\phi}_3 y_{T-3} + \hat{\theta}_1 e_T$$

Point forecasts (h=2)

$$y_t = (1 + \hat{\phi}_1)y_{t-1} - (\hat{\phi}_1 - \hat{\phi}_2)y_{t-2} - (\hat{\phi}_2 - \hat{\phi}_3)y_{t-3} - \hat{\phi}_3 y_{t-4} + \varepsilon_t + \hat{\theta}_1\varepsilon_{t-1}$$

ARIMA(3,1,1) forecasts: Step 2

$$y_{T+2} = (1 + \hat{\phi}_1)y_{T+1} - (\hat{\phi}_1 - \hat{\phi}_2)y_T - (\hat{\phi}_2 - \hat{\phi}_3)y_{T-1} - \hat{\phi}_3 y_{T-2} + \varepsilon_{T+2} + \hat{\theta}_1\varepsilon_{T+1}.$$

ARIMA(3,1,1) forecasts: Step 3

$$\hat{y}_{T+2|T} = (1 + \hat{\phi}_1)\hat{y}_{T+1|T} - (\hat{\phi}_1 - \hat{\phi}_2)y_T - (\hat{\phi}_2 - \hat{\phi}_3)y_{T-1} - \hat{\phi}_3 y_{T-2}.$$

Prediction intervals

Assuming $\varepsilon_t \sim N(0, \sigma^2)$

95% prediction interval

$$\hat{y}_{T+h|T} \pm 1.96\sqrt{v_{T+h|T}}$$

where $v_{T+h|T}$ is estimated forecast variance.

- $v_{T+1|T} = \hat{\sigma}^2$ for all ARIMA models
- Multi-step prediction intervals for ARIMA(0,0,q):

$$y_t = \varepsilon_t + \sum_{i=1}^{q} \hat{\theta}_i \varepsilon_{t-i}.$$

$$v_{T|T+h} = \hat{\sigma}^2 \left[1 + \sum_{i=1}^{h-1} \hat{\theta}_i^2\right], \qquad \text{for } h = 2, 3, \ldots.$$

Prediction intervals

- Prediction intervals increase in size with forecast horizon.
- Prediction intervals can be difficult to calculate by hand
- Calculations assume residuals are uncorrelated and normally distributed.
- Prediction intervals tend to be too narrow.
  - the uncertainty in the parameter estimates has not been accounted for.
  - model uncertainty has not been accounted for.
  - the ARIMA model assumes historical patterns will not change during the forecast period.
  - the ARIMA model assumes uncorrelated future errors.

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Seasonal ARIMA models

ARIMA	$\underbrace{(p, d, q)}$	$\underbrace{(P, D, Q)_m}$
	$\uparrow$	$\uparrow$
	Non-seasonal part of the model	Seasonal part of of the model

- m = number of observations per year.
- d first differences, D seasonal differences
- p AR lags, q MA lags
- P seasonal AR lags, Q seasonal MA lags

Seasonal and non-seasonal terms combine multiplicatively

Seasonal ARIMA models

E.g., ARIMA$(1,1,1)(1,1,1)_4$ model (without constant)

$$(1-\phi_1 B)(1-\Phi_1 B^4)(1-B)(1-B^4)y_t = (1+\theta_1 B)(1+\Theta_1 B^4)\varepsilon_t.$$

- $(1-\phi_1 B)$: Non-seasonal AR(1)
- $(1-\Phi_1 B^4)$: Seasonal AR(1)
- $(1-B)$: Non-seasonal difference
- $(1-B^4)$: Seasonal difference
- $(1+\theta_1 B)$: Non-seasonal MA(1)
- $(1+\Theta_1 B^4)$: Seasonal MA(1)

Seasonal ARIMA models

The seasonal part of an AR or MA model will be seen in the seasonal lags of the PACF and ACF.

ARIMA(0,0,0)(0,0,1)$_{12}$ will show:

- a spike at lag 12 in the ACF but no other significant spikes.
- The PACF will show exponential decay in the seasonal lags; that is, at lags 12, 24, 36, ...

ARIMA(0,0,0)(1,0,0)$_{12}$ will show:

- exponential decay in the seasonal lags of the ACF
- a single significant spike at lag 12 in the PACF.

How does ARIMA() work for seasonal models?

A seasonal ARIMA process

$$\Phi(B)\phi(B)(1-B)^d(1-B)^D y_t = c + \Theta(B)\theta(B)\varepsilon_t$$

Need to select appropriate orders: d, D, p, q, P, Q, and whether to include the intercept c.

Hyndman and Khandakar (JSS, 2008) algorithm:

- Select no. differences d via KPSS test and D using seasonal strength.
- Select p, q, P, Q and c by minimising AICc.
- Use stepwise search to traverse model space.

Box-Jenkins Method

- Suppose that now we get a time series. How can we “reverse-engineer” its functional form or data generating process (DGP)?
  - For example, we expect that an observed time series can be written as

$$y_t = \phi_1 \overbrace{y_{t-1}}^{\text{AR process}} + \phi_{12} \underbrace{y_{t-12}}_{\text{seasonality}} + \overbrace{t\beta_1 + \beta_0}^{\text{deterministic trend}} + \underbrace{\varepsilon_t}_{\text{white noises}}$$

  - How can we discover which term should include in this functional form?
  - How can we identify the coefficients ϕ_1, ϕ_{12}, t?

Box-Jenkins Method

- The Box-Jenkins Method provides a general guideline to diagnose time series. It assumes that time series are composed by multiple temporal processes (e.g. trend, seasonality, autoregressive process, etc.). It then performs diagnostics to compare the observed series with generic forms to decide what processes occur in the data (i.e. DGP).
  - Step 1: Study generic forms and properties
  - Step 2: Study these realizations for an indication of which possibly applies to your data
  - Step 3: Assess your guess-diagnose and iterate
  - Step 4: Perform a meta-analysis at the end to determine which specification is the best